

MR aortography and serum cholesterol levels in patients with long-term nonspecific lower back pain.

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STUDY DESIGN: A cross-sectional analysis of the feeding arteries of the lumbar spine and cholesterol levels on patients with long-term nonspecific lower back pain. OBJECTIVES: To evaluate whether occlusion of lumbar and middle sacral arteries or serum cholesterol levels are associated with lower back pain and/or with disc degeneration. SUMMARY OF BACKGROUND DATA: Atherosclerosis in the wall of the abdominal aorta usually develops at the ostia of branching arteries and the bifurcation, and may obliterate orifices of lumbar and middle sacral arteries. Obstruction of these arteries causes ischemia in the lumbar spine and may result in back symptoms and disc degeneration. METHODS: MR aortography and cholesterol blood tests were performed on 51 patients with long-term lower back pain without specific findings (i.e., spinal or nerve root compression) in regular lumbar MR images. The patients ranged from 35 to 70 years of age (mean age, 56 years). Serum cholesterol and low-density lipoprotein (LDL) cholesterol levels were measured. To assess symptoms and disability NASS low back Outcome Instrument was used. RESULTS: Twenty-nine (78%) of 37 men and 11 (77%) of 14 women showed occluded lumbar and/or middle sacral arteries. The prevalence of occluded arteries was 2.5 times more than in subjects of corresponding age group in a Finnish necropsy material. Twenty-three (62%) men and seven (50%) women had significant disc degeneration. Disc degeneration was associated with occluded lumbar/middle sacral arteries ($P = 0.035$). Patients with occluded arteries or significant disc degeneration did not complain more severe symptoms than those without, whereas patients with above normal serum LDL cholesterol scored higher in neurogenic symptoms ($P = 0.031$) and complained more often severe pain ($P = 0.049$) than those with normal LDL cholesterol. CONCLUSIONS: The study indicates that lumbar and middle sacral arteries are often occluded in patients with nonspecific long-term lower back pain. Occlusion of these arteries may also be associated with disc degeneration.